

Mark schemes

Q1.

- (a) for $-5 \leq x \leq 15$

$$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} \left(\frac{x+5}{20} \right) = \frac{1}{20}$$

AG

B1

1

- (b) (i) $P(X \geq 7) = 1 - F(7) = 1 - \frac{12}{20} = \frac{2}{5}$ or $\left[\frac{8}{20}; \frac{4}{10}; 0.4 \right]$

Alternative:

Use of $f(x) = \frac{1}{20}$ or graph $\Rightarrow$

$$P(X \geq 7) = \frac{1}{20} \times (15 - 7) = \frac{2}{5} \text{ (oe)}$$

B1

1

- (ii) $P(X \neq 7) = 1$
cao

B1

1

- (iii) $E(X) = \frac{1}{2} (-5 + 15) = 5$

Alternative:

$$E(X) = \int_{-5}^{15} \frac{x}{20} dx = \left[\frac{x^2}{40} \right]_{-5}^{15} = \frac{1}{40} (225 - 25) = \frac{1}{40} \times 200 = 5$$

B1 (cao)

B1

1

- (iv) $E(3X^2) = \int_{-5}^{15} \frac{3x^2}{20} dx$ (ignore limits)

M1

$$\left. \begin{aligned} & \left[\frac{x^3}{20} \right]_{-5}^{15} \\ & \frac{1}{20} (3375 + 125) \\ & 168\frac{3}{4} + 6\frac{1}{4} \end{aligned} \right\}$$



A1

$$= 175$$

(cao) (allow 174.9)

A1

3

Alternative:

$$\text{Var}(X) = \frac{1}{12}(15 - -5)^2 = \frac{400}{12} \text{ (oe)}$$

$$E(3X^2) = 3 E(X^2)$$

(B1)

$$E(3X^2) = 3 \times \left[\frac{400}{12} + 5^2 \right]$$

$$= 3 \times \left[\{ \text{their Var}(X) > 0 \} + \{ \text{their } E(X) \}^2 \right] \text{ used } (\Rightarrow M1)$$

(M1)

$$= 175$$

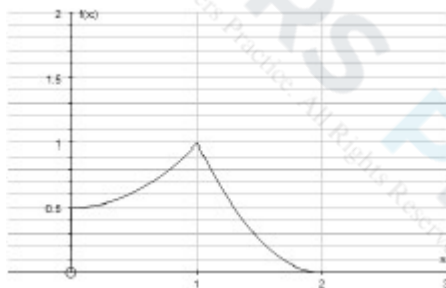
(A1)

(3)

[7]

Q2.

(a)



B1 for axes

B1 for curve from (0, 0.5) to (1, 1)

B1 for curve from (1, 1) to (2, 0)

B3

3

(b)

$$P(X \leq 1) = \int_0^1 \frac{1}{2}(x^2 + 1) dx$$

M1

$$= \left[\frac{x^3}{6} + \frac{x}{2} \right]_0^1$$

A1

$$= \left[\frac{1}{6} + \frac{1}{2} \right] = \frac{2}{3}$$

0.667

A1

3

(c) $E(X^2) = \int_0^1 x^2 \times \frac{1}{2}(x^2 + 1) dx + \int_1^2 x^2 (x - 2)^2 dx$

both integrals seen

M1

$$= \left[\frac{x^5}{10} + \frac{x^3}{6} \right]_{x=0}^{x=1} + \left[\frac{x^5}{5} - x^4 + \frac{4x^3}{3} \right]_{x=1}^{x=2}$$

A1A1

$$= \left(\frac{1}{10} + \frac{1}{6} \right) + \left(\left[\frac{32}{5} - 16 + \frac{32}{3} \right] - \left[\frac{1}{5} - 1 + \frac{4}{3} \right] \right)$$

dep(M1)

m1

$$= \frac{4}{5}$$

AG

A1

5

(d) (i) $E(X) = \frac{19}{24}$ and $k\text{Var}(X) = 499$
 $\text{Var}(X) = E(X^2) - E^2(X)$
 $= \frac{4}{5} - \left(\frac{19}{24} \right)^2$

M1

$$= \frac{499}{2880} \quad (0.173)$$

A1



$$\Rightarrow k = 2880$$

CAO

A1

3

$$\begin{aligned} \text{(ii)} \quad E(5X^2 + 24X - 3) \\ = 5E(X^2) + 24E(X) - 3 \end{aligned}$$

M1

$$\begin{aligned} &= 5 \times \frac{4}{5} + 24 \times \frac{19}{24} - 3 \\ &= 20 \end{aligned}$$

CAO

A1

2

$$\text{(iii)} \quad \text{Var}(12X - 5) = 144\text{Var}(X)$$

M1

$$\begin{aligned} &= 144 \times \frac{499}{2880} \\ &= \frac{499}{20} \text{ or } (24.95) \\ &\text{CAO (AWFW 24.9 to 25)} \end{aligned}$$

A1

2

[18]

Q3.

$$\text{(a) (i)} \quad E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} 3x^2 \, dx = \int_0^1 3x \, dx$$

M1

$$= \left[\frac{3x^2}{2} \right]_0^1 = 1.5$$

CAO

A1

A1

3

$$\text{(ii)} \quad E\left(\frac{1}{X^2}\right) = \int_0^1 \frac{1}{x^2} 3x^2 \, dx = \int_0^1 3 \, dx$$

M1

$$= [3x] = 3.0$$

A1

$$\text{Var}\left(\frac{1}{X}\right) = 3.0 - (1.5)^2$$

dependent on previous M1

m1

$$= 0.75$$

[on their (i)] and Var > 0

A1ft

4

(b)
$$E\left(\frac{5+2X}{X}\right) = E\left(\frac{5}{X} + 2\right)$$

M1

$$= 5E\left(\frac{1}{X}\right) + 2$$

M1

$$= 5 \times 1.5 + 2$$

$$= 9.5$$

CAO

A1

$$\text{Var}\left(\frac{5+2X}{X}\right) = \text{Var}\left(\frac{5}{X} + 2\right)$$

$$= 25 \times \text{Var}\left(\frac{1}{X}\right)$$

M1

$$= 25 \times 0.75$$

$$= 18.75$$

CAO

A1

5

[12]

Q4.

(a) $E(X) = 4 \quad E(X^2) = 17.2$



$$\text{Var}(X) = E(X^2) - (E(X))^2$$

use of

M1

$$= 17.2 - 42$$

$$= 1.2$$

CAO

A1

2

(b) $C = 2\pi(X + 8)$

OE
Either

or $E(X + 8) = 12$

CAO

B1

Thus $E(C) = 24\pi$

CAO

B1

$$\text{Var}(C) = 4\pi^2 \times \text{Var}(X)$$

*use of $V(aX + b) = a^2 \times V(X)$
with $a > 1$ and $b > 0$*

M1

Thus $\text{Var}(C) = 4.8\pi^2$

ft on $V(X)$; but must include π

A1ft

4

(c) (i) Area, $S = \pi(X + 8)^2$

use of πr^2

M1

$$= \pi(X^2 + 16X + 64)$$

Thus $a = 16$ and $b = 64$

CAO both

A1

2

(ii) $E(S) = \pi(E(X^2) + 16E(X) + 64)$

*attempted application of E to expanded expression in (c)(i)
M0 for use of $E(X^2) = 42 = 16$*

M1

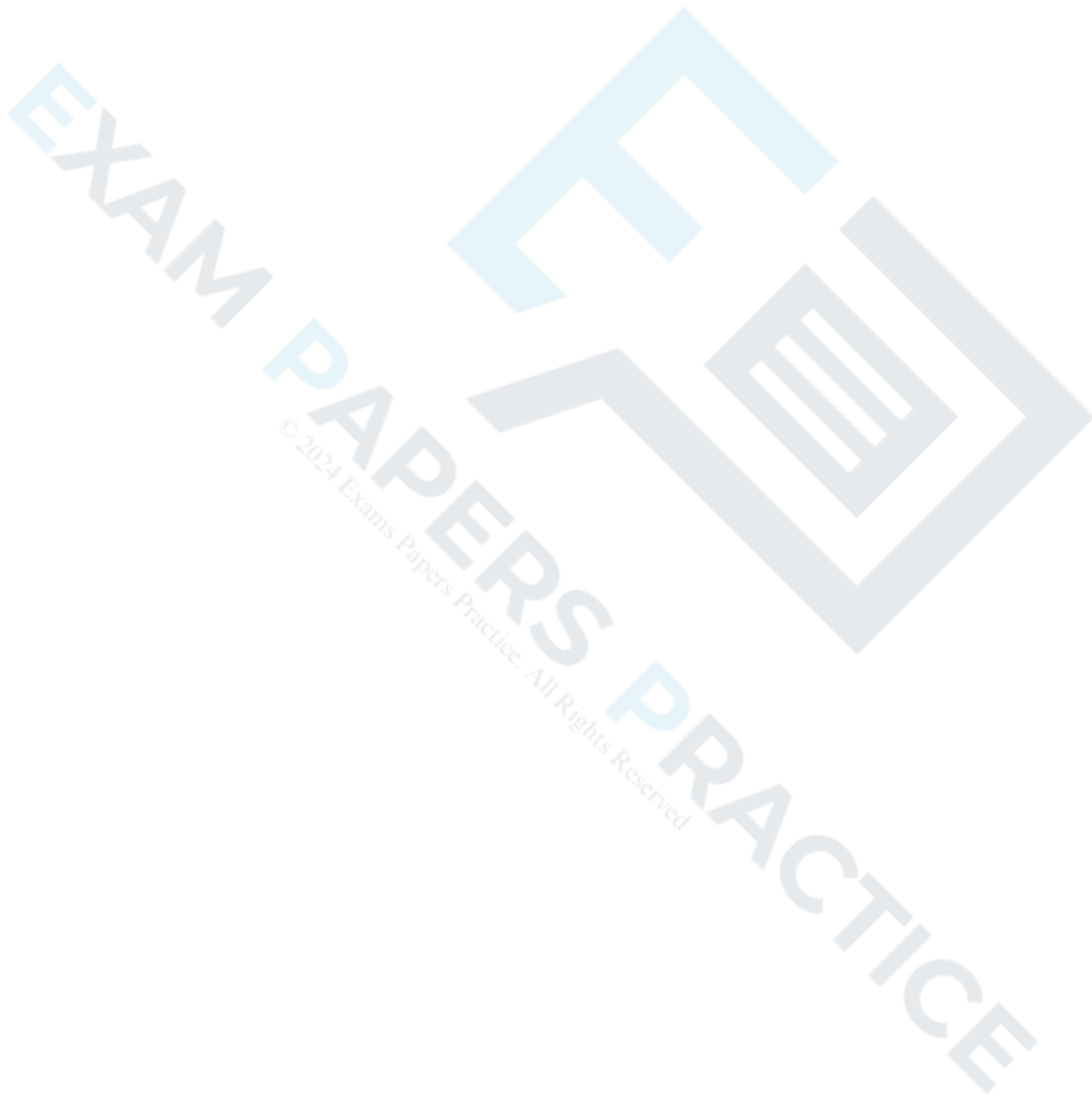
$$= \pi(17.2 + 64 + 64) = 145.2\pi$$

CAO

A1

2

[10]





Examiner reports

Q1.

In part (a), most students were able to indicate that the differentiation of $F(x)$ was required even though use of the correct notation was often sadly lacking. In part (b), those who realised that they were dealing with a continuous *rectangular* distribution, found parts (i), (ii) and (iii) very easy; simply writing down the required answers. Unfortunately, in part (b)(i), some students thought that the answer was 0.6 whilst others assumed a discrete distribution and consequently attempted to find $1 - P(X \leq 6)$. In part (b)(ii), a common incorrect answer was $P(X \leq 7) = 0$. In part (b)(iii), the correct answer $E(X) = 5$ was usually

seen but most students, unnecessarily, used calculus to evaluate $\int_{-3}^{13} \frac{x}{20} dx$ thus wasting time.

In part (b)(iv), the great majority chose to correctly evaluate $\int_{-3}^{13} \frac{3x^2}{20} dx$ to achieve the required answer of 175.

Q2.

In the previous Report on the Examination, it stated that too many candidates seemed to be under the misapprehension that a request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram taken from the display on their graphics calculators. This still seemed to be the case and all such attempts lost some or all of the marks available. The axes should have been drawn with the aid of a ruler and pencil and, as a minimum, should have contained the appropriate scales; $O(0, 0)$, $(1, 0)$ and $(2, 0)$ on the x -axis and $(0, 0.5)$ and $(0, 1)$ on the y -axis. The two concave arcs were often seen as being convex or as straight lines.

In part (b), there were many correct solutions with $\int_0^1 \frac{1}{2}(x^2 + 1)dx = \frac{2}{3}$ often seen. Unfortunately, those candidates who drew straight lines in their sketches, usually ended up with $P(X \leq 1) = \frac{1}{2}(0.5 + 1) = \frac{3}{4}$; this gained no credit.

Part (c) was answered very well by many candidates with the standard of algebra and integration showing a marked improvement. However, it should be noted that, where an answer is given on the examination paper, sufficient method must be shown to ensure that full marks are awarded.

Part (d)(i) was answered very well by many candidates. However, some candidates,

having found that $\text{Var}(X) = \frac{4}{5} - \left(\frac{19}{24}\right)^2 = \frac{499}{2880}$, then were unable to write down the value of k as 2880. Whilst most candidates obtained the correct answer in part (d)(ii), a minority

thought incorrectly that $E(X^2) = \left(\frac{19}{24}\right)^2$ even though the correct value of $\frac{4}{5}$ was given in the question. Part(d)(iii) was quite well attempted although some candidates used $\text{Var}(12X - 5) = 12 \times \text{Var}(X)$ or even $\text{Var}(12X - 5) = 12^2 \times \text{Var}(X) - 5$.

Q3.



This was the least-well-answered question on the paper with many major errors of technique and understanding. It was only answered well by the more able candidates. It was a major concern that so many candidates thought incorrectly that

$E\left(\frac{1}{X}\right) = \frac{1}{E(X)}$ and $\text{Var}\left(\frac{1}{X}\right) = \frac{1}{\text{Var}(X)}$. These candidates wasted a lot of time evaluating $E(X)$ and $E(X^2)$ to no avail. There were also many examples of poor integration

techniques. When $E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} \times 3x^2 dx$ was correctly stated, some candidates then

$$\frac{3x^2}{3} \text{ or } \frac{3x^3/3}{x^2/2}$$

seemed unable to integrate correctly, with $\ln x \times$ often seen.

The fact that the variance cannot be less than zero seemed to have escaped the attention of some candidates who were apparently quite comfortable, not only in obtaining negative values for their variance, but also in using such values in part (b), where it was also evident that many candidates were let down by poor algebraic skills. It was expected that

candidates would first express $\frac{5+2X}{X}$ as $\frac{5}{X} + 2$ and then use

$E\left(\frac{5}{X} + 2\right) = 5E\left(\frac{1}{X}\right) + 2$ and $\text{Var}\left(\frac{5}{X} + 2\right) = 5^2 \text{Var}\left(\frac{1}{X}\right)$ to obtain answers of 9.5 and 18.75 respectively. Unfortunately many candidates could not even cope with the first step. On the other hand, there were some good attempts from evaluating

$E\left(\frac{5+2X}{X}\right)$ and $E\left(\left(\frac{5+2X}{X}\right)^2\right)$ by integration.

Q4.

Answers to part (a) were usually correct, although 4^2 was sometimes omitted or 4 was not squared. Some candidates could apparently not recall the formula for the circumference of a circle. Those candidates who could, invariably obtained 24π for its mean. The variance caused more trouble, often because π was not squared, or was omitted. In other solutions, the 2 was not squared. The correct use of brackets in expanding $(2\pi)^2$ seemed to have been forgotten by far too many candidates. Part (c)(i) was often answered correctly, but again there were some poor attempts at the expansion of $(X+8)^2$. Part (c)(ii) caused considerable difficulty for many candidates because they were under the mistaken impression that $E(X^2)$ was the same as $(E(X))^2$. Hence they obtained 144π , rather than 145.2π , for which no marks were awarded.